Indian Institute of Technology Kharagpur
B.Tech Project · Vinod Gupta School of Management

MULTIMODAL DEEP LEARNING FOR MARKET PREDICTION

How Much Do Evaluation Choices Inflate Multimodal Market Prediction, and Does News Help Under Market Stress?

An audit of price, FinBERT news and volatility-regime models on the NIFTY / SPY benchmark: 3 protocols, 8 models, 10 seeds.

Price window
10 days × 12 features

FinBERT headlines
768-d or 3 scores

Volatility regime
k-means, k = 3

↓ Fall · Neutral · Rise ↓

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02 · ANSWER FIRST

News embeddings hurt, sentiment is neutral, and protocol effects are small and model-specific

0.466

Price-only LSTM

Strict-protocol Macro-F1; random-guess floor 0.333

−0.050

Add FinBERT embedding

95% interval [−0.087, −0.012]: reliably worse

+0.016

Add 3 sentiment scores

Interval [−0.017, +0.049] includes zero

3 of 12

Protocol gaps exclude zero

Regime-only M5 gains; M6 falls under P1

RQ1 · EVALUATION PROTOCOL

Looser protocols do **not** systematically inflate scores. P1 also changes the test sample, so gaps mix protocol and sample effects.

RQ2 · VALUE OF NEWS BY REGIME

No reliable news gain overall or in the 50-row 2020 stress window. One Mid-vol sentiment interval excludes zero and is treated as a hypothesis.

FINANCE

A public benchmark with a chronological split and a market shock inside the test period

OFFICIAL CHRONOLOGICAL SPLIT (NOT TO SCALE)



2010-01-06 to 2017-06-27 (15 warm-up rows dropped) to 2019-02-12 to 2020-09-21

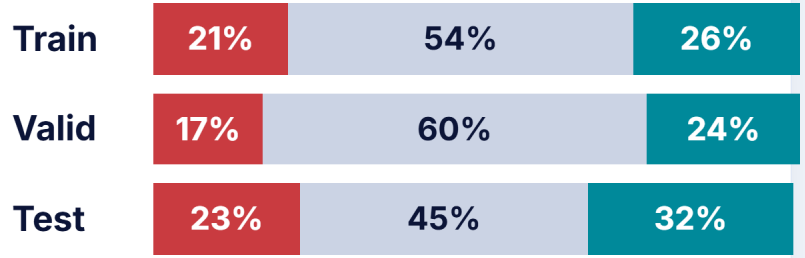
Stress window 1 Feb to 30 Apr 2020: **50 test rows**, inside the test period

2,096 rows
1,462 / 317 / 317 after dropping 15 warm-up rows

10 × 12
window: days × scale-free features

±0.5%
next-day close-to-close bands

LABEL MIX SHIFTS BY SPLIT



Fall · Neutral · Rise. Majority-class Macro-F1 is only 0.207.

HEADLINES PER DAY AFTER THE 50 CAP

49.0
train
raw 277

46.4
valid
raw 54

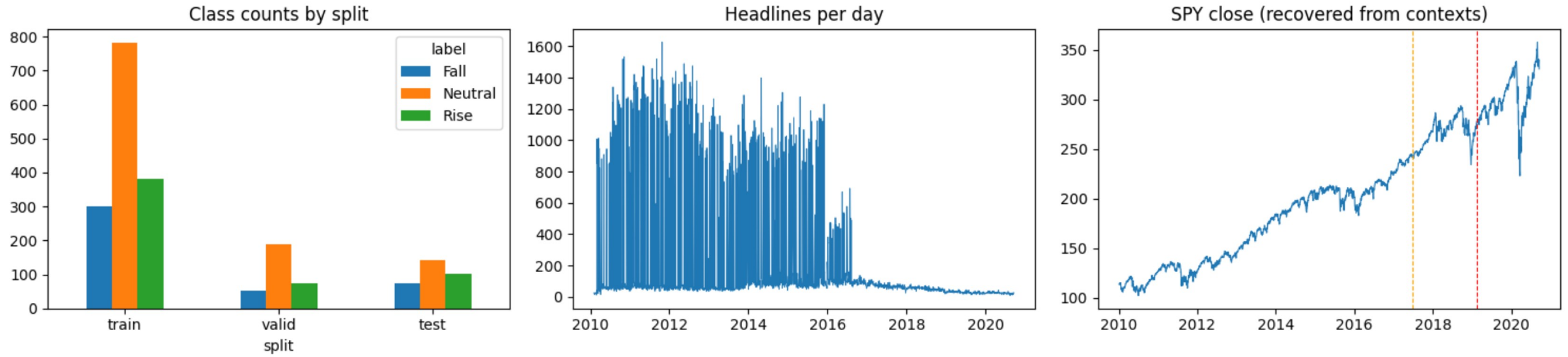
26.2
test
raw 26

Source: NIFTY (Saqur et al., 2024); results/logs/01 and 03. 94,652 headlines embedded.

FINANCE STATS

Both the label mix and the news supply change between eras

PRICE HISTORY, LABEL MIX AND NEWS VOLUME BY SPLIT



Label shift

Neutral is 54% of training days, 60% in validation and 45% in test; Rise rises from 26% to 32%. Accuracy hides this, Macro-F1 does not.

News supply shift

About 277 raw headlines a day in training, 54 in validation and 26 in test; the 50 cap leaves 49 / 46 / 26.

Tone shift

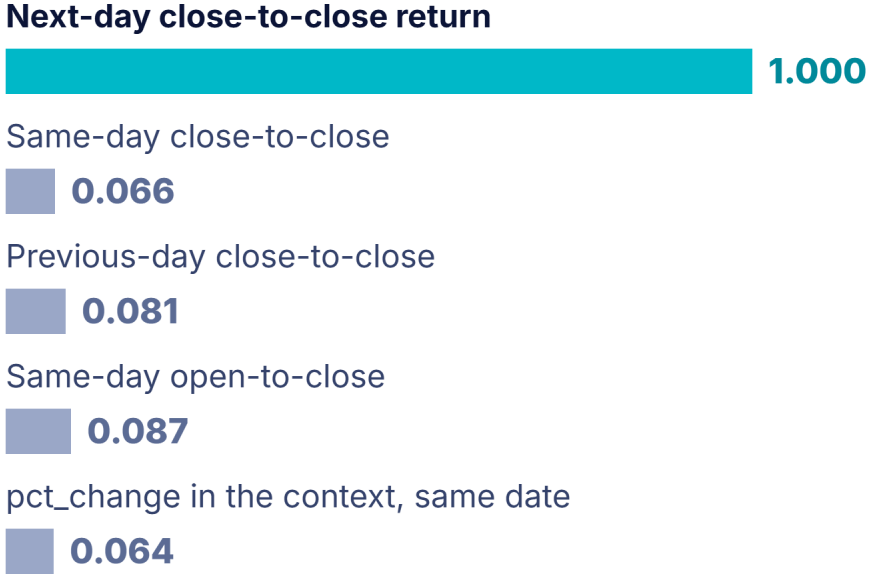
Mean FinBERT sentiment −0.068, −0.095 and −0.184 by split; March 2020 reaches −0.319 against −0.176 for the rest of test.

Source: report Figure 3; results/logs/01b_eda.txt, 03_finbert_embeddings.txt.

FINANCE ML

The label is the next trading day’s return: verified, then enforced in code

SHARE OF ROWS MATCHING THE LABEL (±0.001)



Verified on 2,110 comparisons; the last test row has no next-day close in the recovered series.

LAG = 0 TIMING CONVENTION



A 10-day window ends at **t**; the label is the return from t to t+1. Nothing after t is an input.

GUARDS AND AUDIT RESULTS

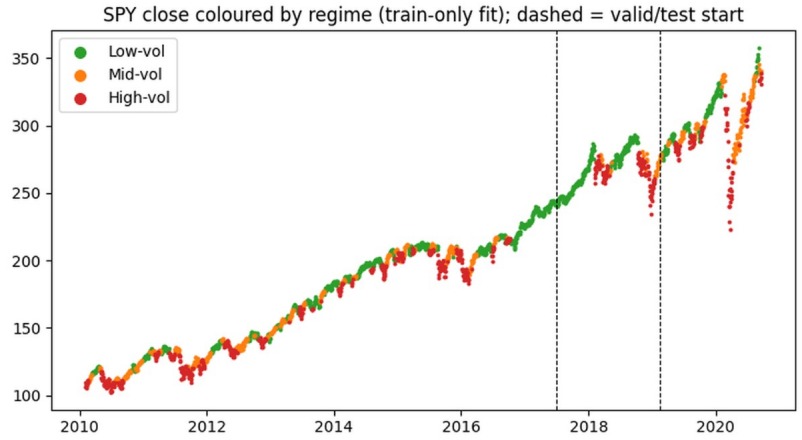
- Every window and regime feature is asserted to use no day after the row’s date.
- 2,695 trading days recovered from overlapping contexts; duplicates agree exactly.
- One trading day (2018-12-31) is absent; the other 14 flagged dates are NYSE closures.
- 10 of 2,111 labels sit on the ±0.5% boundary, most likely rounding (kept).

Source: results/logs/01_data_load_and_timing_audit.txt; report Section 3.2 to 3.3.

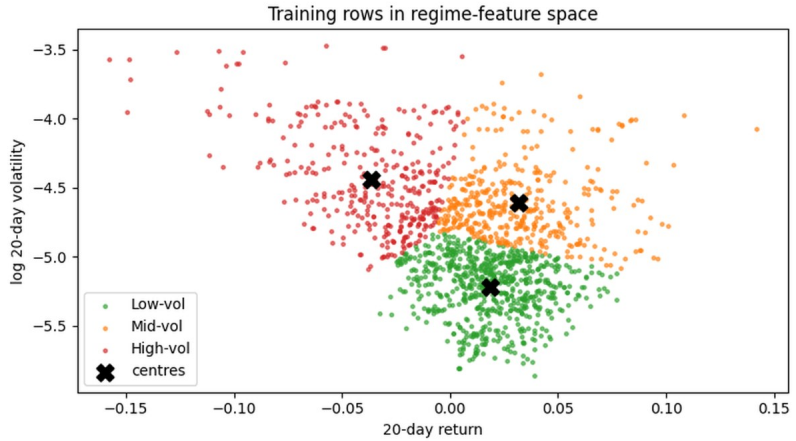
FINANCE ML

Regimes: Low-vol persists, Mid- and High-vol flicker

SPY COLOURED BY REGIME (TRAIN-ONLY FIT)



TRAINING ROWS IN REGIME-FEATURE SPACE



Low-vol — 47.9% of train

0.55% 20d vol · **+1.9%** 20d return · **0.00%** next day

median run **15 days** · stay probability **0.961**

Next day: Fall · Neutral · Rise

16%	64%	19%

Mid-vol — 28.1% of train

1.03% 20d vol · **+3.2%** 20d return · **+0.09%** next day

median run **4 days** · stay probability **0.869**

Next day: Fall · Neutral · Rise

18%	55%	27%

High-vol — 24.0% of train

1.26% 20d vol · **−3.6%** 20d return · **+0.10%** next day

median run **4 days** · stay probability **0.875**

Next day: Fall · Neutral · Rise

31%	29%	41%

Source: results/logs/02_regimes.txt; report Section 4.4. Label mix is over all splits (strict fit).

ML

One small network with switchable inputs gives six comparable learned models

Price window
10 trading days × 12 scale-free features

→ **LSTM, 64 units**
last hidden state, dropout 0.2

Headlines
≤ 50 per day, fixed seed 1234

→ **Frozen FinBERT → MLP**
768-d mean pool → Linear 768→64, ReLU (M6: 3 sentiment scores)

Regime features
20-day return, log 20-day volatility

→ **k-means → embedding**
3 clusters → 8-d vector

→ **CONCATENATE THE ACTIVE BRANCHES**

- **M1** price
- **M2** price + FinBERT
- **M3** M2 + regime
- **M4** FinBERT only
- **M5** regime only
- **M6** price + sentiment

→ **MLP head**
Linear → 64, ReLU, dropout, Linear 64 → 3

↓

Fall · Neutral · Rise
next-day close-to-close move, ±0.5% bands

Adam · lr 1e-3 · weight decay 1e-4

batch 64 · ≤ 50 epochs · patience 8

class-weighted cross-entropy · clip 5

baselines: M0 majority · M0r random

single-factor ablations: one input changes per comparison

same recipe for every model, no per-model tuning

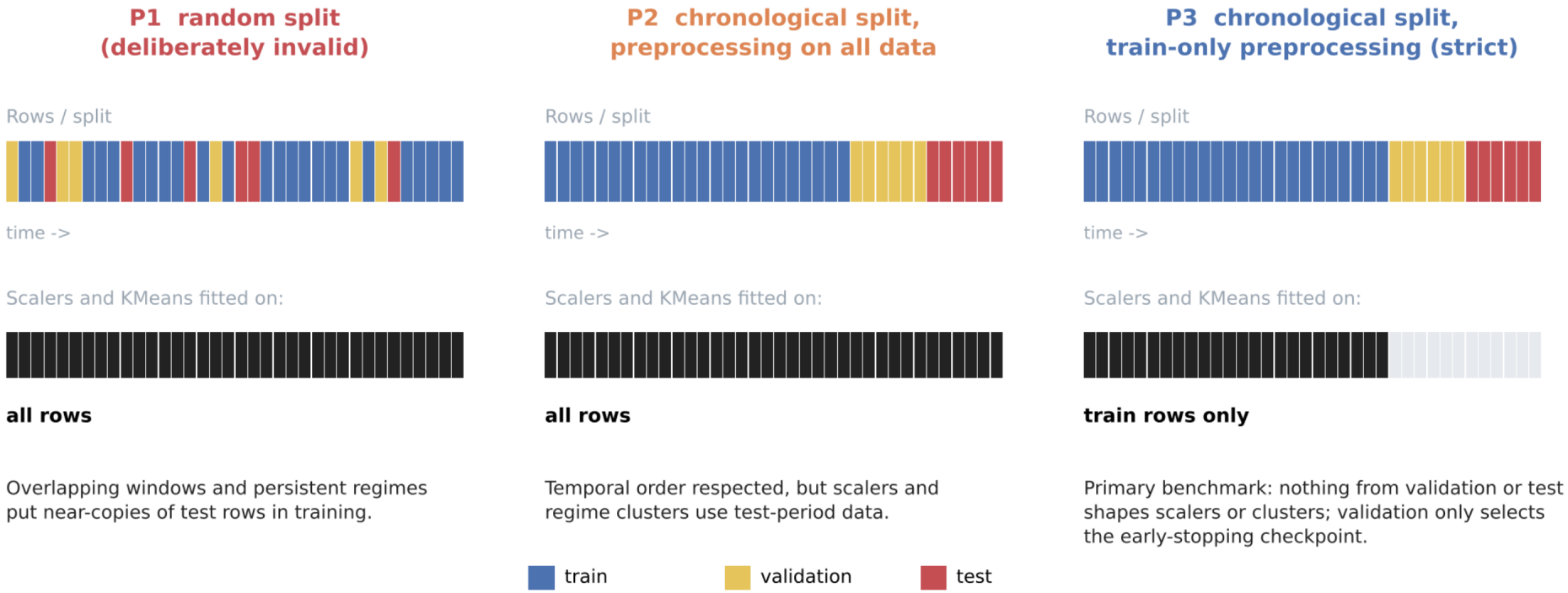
frozen encoder, 10 seeds

checkpoint picked on validation only

Source: report Figure 1 and Table 6; src/04_models.py. A gated variant (M7) is implemented but was not run.

ML FINANCE

Three protocols run on identical models and seeds



P1 – P3: look-ahead through shuffled neighbours and global fitting; test sample differs

P2 – P3: look-ahead via global scalers and clusters only; same 317 test rows

P3: point-in-time, as a live system would see it. Primary

Source: report Figure 5 and Section 4.7. 8 models × 3 protocols × 10 seeds = 240 evaluations (180 trained, 60 baseline).

STATS FINANCE

Macro-F1 is the primary metric, and every effect gets an interval

TEST METRICS UNDER P3 (MEAN OF 10 SEEDS)

		ACCURACY	BAL. ACC.	MACRO-F1
M0	Majority class	0.451	0.333	0.207
M0r	Uniform random	0.339	0.341	0.333
M1	Price LSTM	0.530	0.476	0.466
M2	+ FinBERT	0.493	0.448	0.416
M6	+ sentiment	0.531	0.498	0.482

Always predicting the majority class gets **0.451 accuracy** but **0.207 Macro-F1** and balanced accuracy of exactly 1/3. Accuracy alone would reward it.

LEAKAGE IN FINANCE TERMS

P1: look-ahead through neighbours
Shuffled days: a test day sits between training days and its near-identical window features leak. Global scalers and clusters add more.

P2: look-ahead through preprocessing
Time order kept, but scalers and k-means see future rows.

P3: point-in-time
Everything fitted on the past only, as a live system would. Checkpoint chosen on validation.

10 **seeds per cell**
180 trained + 60 baseline runs

2,000 **seed-bootstrap draws**
protocol-gap intervals

1,000 **row-bootstrap draws**
model-difference intervals

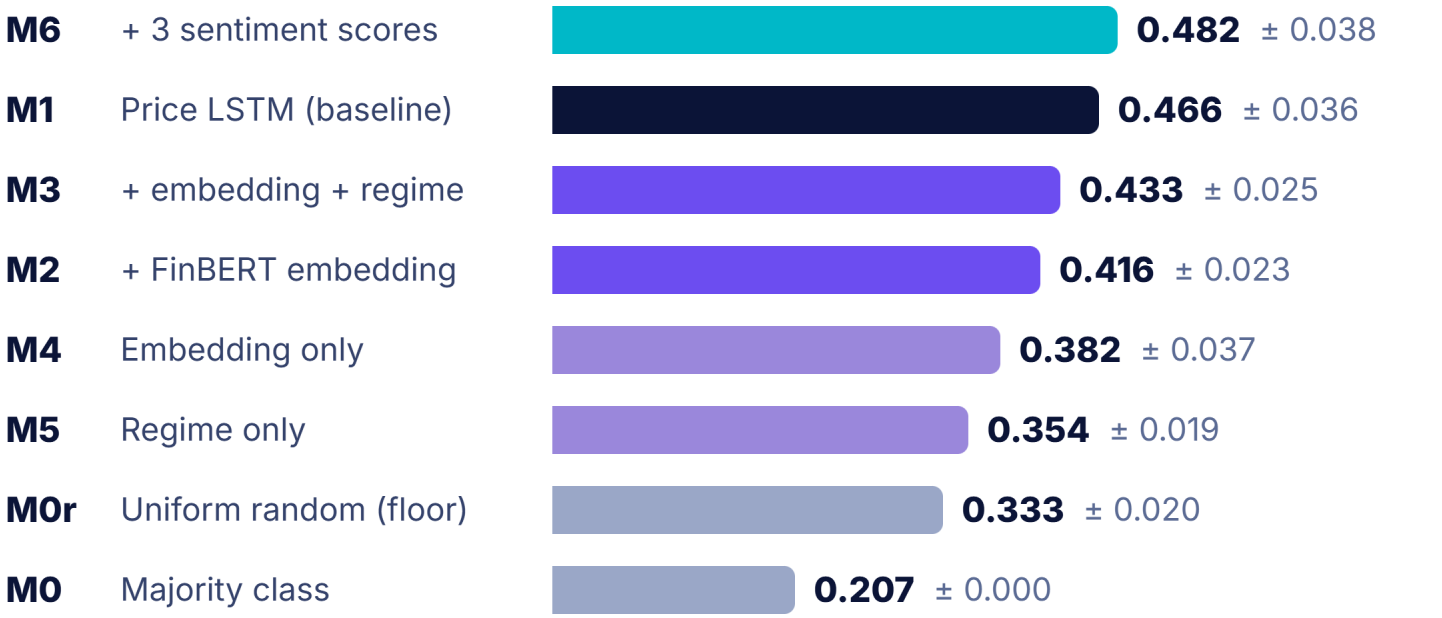
7 of 36 **intervals exclude 0**
~1.8 expected by chance

Source: EXECUTED_RESULTS.md; report §4.6, §4.8. Chance count assumes independent 5% tests (illustrative).

ML

Under the strict protocol, a price-only LSTM is already hard to beat

TEST MACRO-F1, STRICT PROTOCOL P3 (MEAN ± S.D., 10 SEEDS)



Majority-class accuracy is 0.451 but its Macro-F1 is 0.207, which is why Macro-F1 is primary. M6 was added after a pilot, so its edge is unconfirmed.

−0.050

M2 − M1

95% CI [−0.087, −0.012]. Embedding is reliably worse.

+0.016

M6 − M1

CI [−0.017, +0.049]. Includes zero.

+0.017

M3 − M2 (regime input)

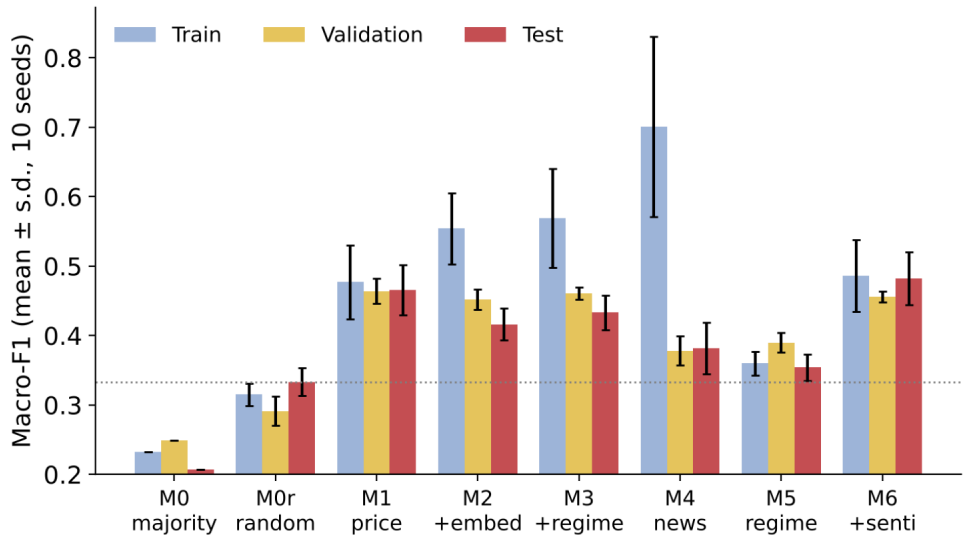
CI [−0.002, +0.038]. H3 not supported.

Source: results/EXECUTED_RESULTS.md. Test set = 317 rows, 2019-02-13 to 2020-09-21.

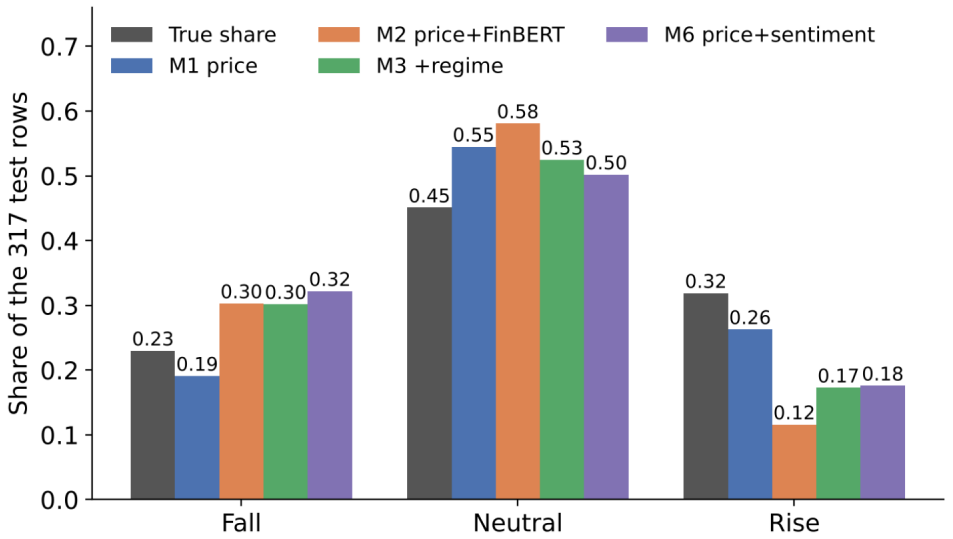
ML

The embedding models memorise training rows and under-predict Rise

TRAIN, VALIDATION AND TEST MACRO-F1 (P3)



PREDICTED VS TRUE CLASS SHARES (TEST)



Capacity
768 inputs for ~1,460 training rows: M4 hits 0.701 on train, 0.382 on test.

Weak daily signal
Cosine 0.963 between consecutive rows vs 0.951 for random pairs.

Era shift
Headlines per day 49 → 26 and mean sentiment −0.07 → −0.18, train to test.

Class skew
News models predict Rise for only 12–18% of rows vs 32% true.

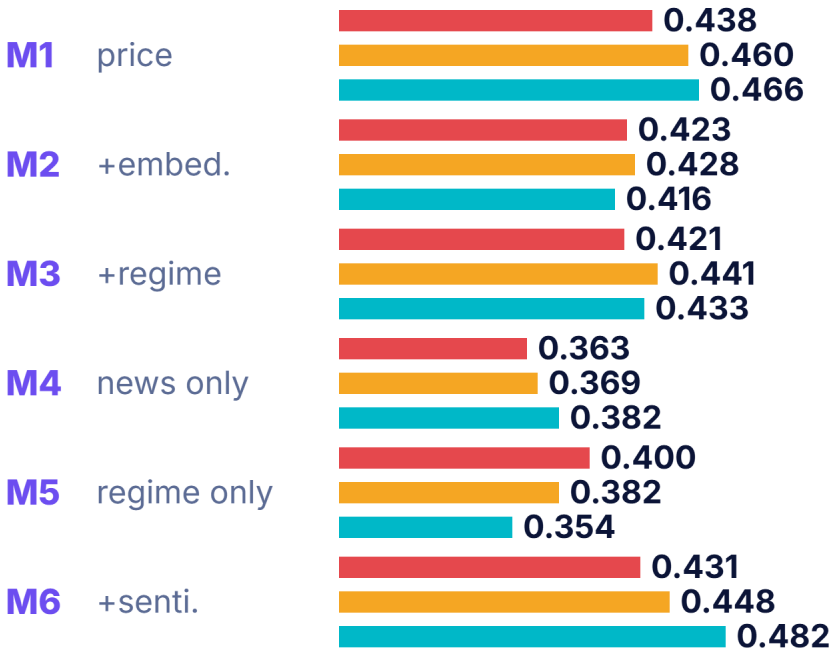
Source: EXECUTED_RESULTS.md §3 and §5; logs 03, 06. Explanations are untested hypotheses.

STATS FINANCE

Looser protocols did not uniformly inflate scores

TEST MACRO-F1 BY PROTOCOL

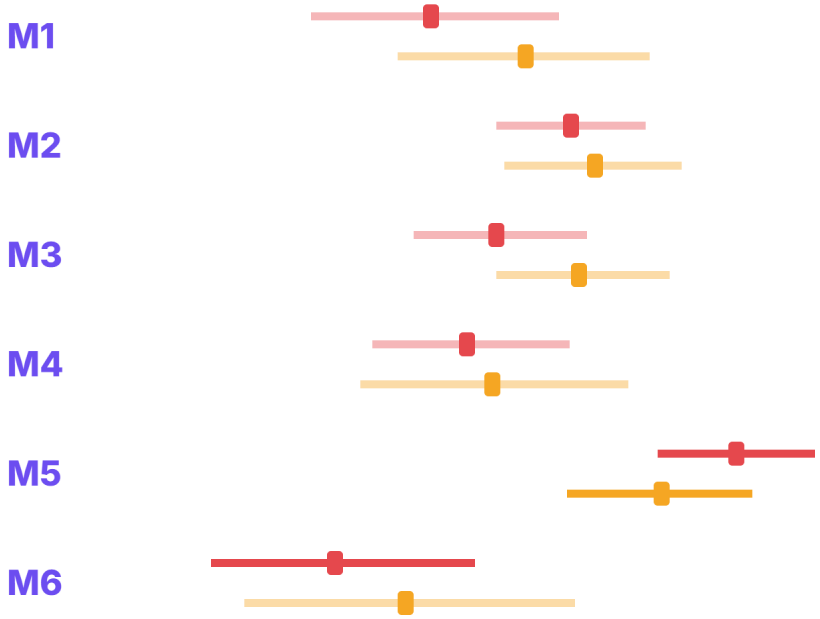
■ P1 random ■ P2 chronological ■ P3 strict



GAP TO P3 WITH 95% INTERVALS (SEED BOOTSTRAP)

— P1 − P3 — P2 − P3 solid = excludes zero

dashed line = zero gap · left: looser protocol lower · right: looser higher



H1 is not supported in general. M5 gains under both looser protocols (+0.046, +0.028) and M6 falls under P1 (−0.051). Mechanisms are not isolated here, and P1 also changes the test sample.

Source: results/audit_gaps.csv; report Tables 10 and 11. Intervals: 2,000 resamples of the 10 per-seed scores.

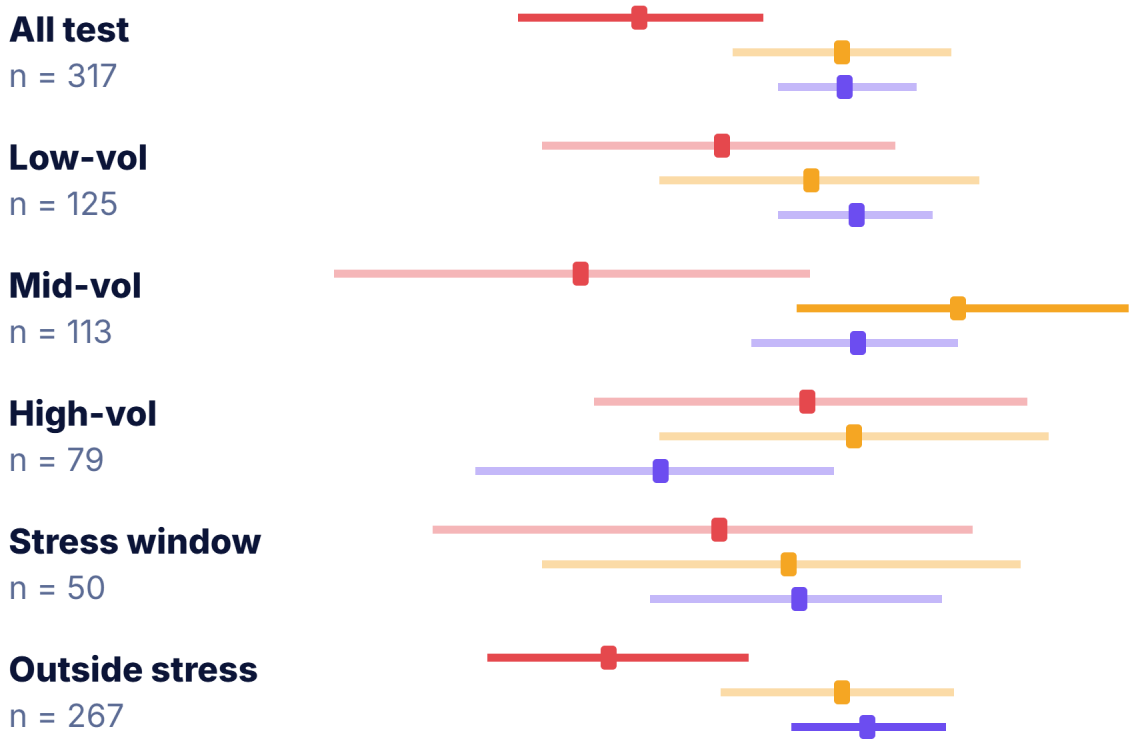
FINANCE STATS

No reliable news gain under the 2020 stress

CHANGE IN MACRO-F1 UNDER P3, 95% ROW-BOOTSTRAP INTERVALS

— M2 − M1 embedding — M6 − M1 sentiment — M3 − M2 regime

solid = excludes zero · scale −0.16 to +0.12



MACRO-F1 BY REGIME GROUP (P3)

	M0r	M1	M2	M3	M6
All test	0.333	0.466	0.416	0.433	0.482
Low-vol	0.329	0.409	0.386	0.407	0.415
Mid-vol	0.322	0.410	0.341	0.361	0.464
High-vol	0.323	0.357	0.363	0.319	0.378
Stress (50)	0.316	0.473	0.448	0.450	0.471
Outside	0.334	0.451	0.391	0.416	0.467

darker = higher · M0r is the random floor

4 of 24

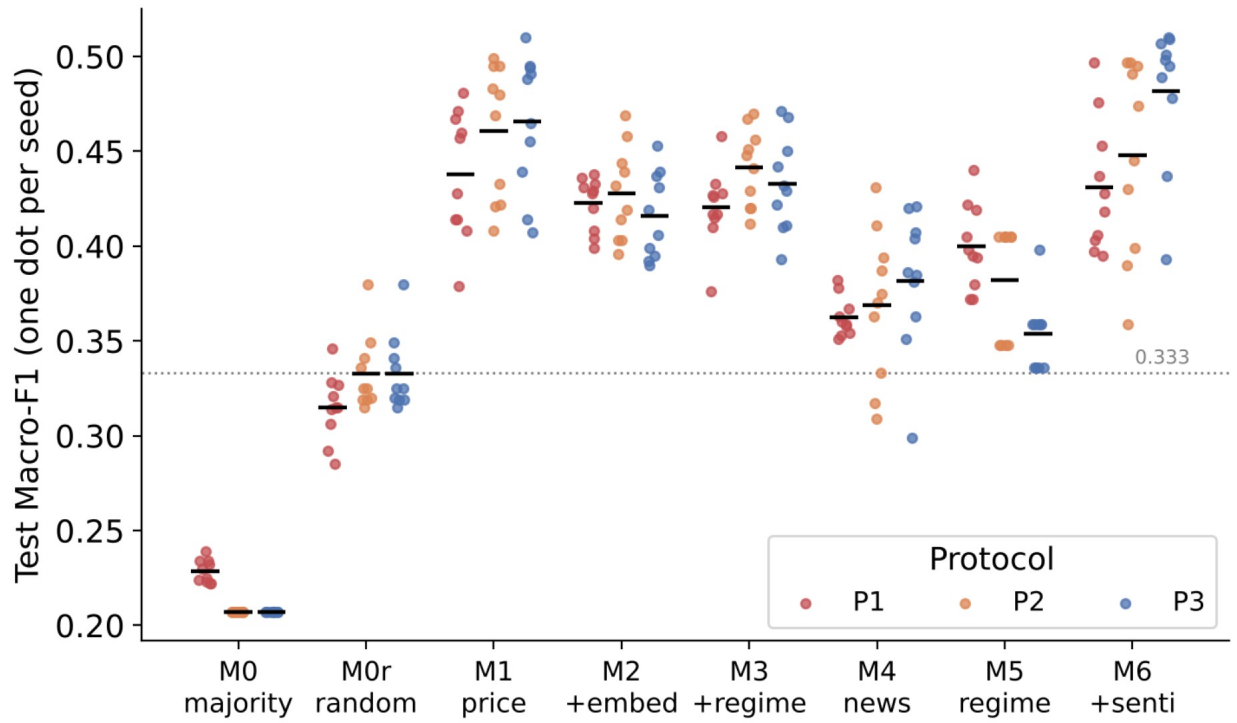
intervals exclude zero: embedding overall and outside stress, sentiment in Mid-vol (+0.054), regime outside stress (+0.024). All exploratory.

Source: results/p3_diffs_by_group.csv; report Table 13. Row bootstrap, 1,000 resamples; temporal dependence ignored.

STATS

Seed spread is as large as most effects, so every claim carries an interval

ALL 240 EVALUATIONS (ONE DOT PER SEED, BAR = MEAN)



HOW INTERVALS ARE BUILT

Protocol gaps: resample the 10 per-seed scores, 2,000 draws; excludes test-row noise.
Model differences: resample test rows, 1,000 draws; treats rows as independent, so intervals are probably too narrow.

LIMITS

- One test period, dominated by the 2020 shock
- 36 intervals tested, 7 exclude zero: exploratory
- M0r and M6 added after a pilot (not saved)
- Headline times unknown; M7 and LAG=1 not run

Source: report Figure 8 and Sections 4.8, 6.5. A clean re-run from raw data has not been independently verified.

15 · CONCLUSION

A strong price baseline, a news input that does not help, and an evaluation that needs intervals

1

Price-only is hard to beat

0.466 Macro-F1 under the strict protocol; the 768-d FinBERT embedding lowers it by 0.050.

2

Protocol effects are model-specific

Only the regime-only model gains under looser protocols; P1 also changes the test sample.

3

No stress-window news edge

Nothing reliable in 50 rows; one Mid-vol sentiment gain is a hypothesis.

FUTURE WORK

Run the regime-gated model M7 and the LAG = 1 window; block-bootstrap intervals; timestamped headlines; more benchmarks and markets.

WHAT THIS IS NOT

An evaluation audit on one benchmark. It is not a trading system and does not establish profitability.

16 · THANK YOU

Questions?

Code, logs and the report: github.com/Hamza-Faarooq/btp-regime-aware-multimodal-forecasting

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DATA AND MODELS

Saqur, Kato, Vinden & Rudzicz (2024). NIFTY Financial News Headlines Dataset. arXiv:2405.09747

Araci (2019). FinBERT. arXiv:1908.10063

Hochreiter & Schmidhuber (1997). Long Short-Term Memory. Neural Computation 9(8)

EVALUATION AND MULTIMODAL FUSION

Bailey, Borwein, López de Prado & Zhu (2014, 2015). Backtest overfitting

Gu et al. (2024) FinBERT-LSTM, arXiv:2407.16150; Karadaş et al. (2025), arXiv:2502.05186

Zong et al. (2024) gated cross-attention fusion, arXiv:2406.06594